An EDLines-Based Virtual Encoder with Pose Estimation for XY-Axis Calibration of a Cartesian 3D Printer

Zayd Zihni Zaidan, *Member, IEEE*, Sylmina Dalily Alkaff, Aqilah Baseri Huddin, Hanim Hussin, Asif Ali Zamzami, *Member, IEEE* and Bahrudin

***Abstract*—In the era of Industry 4.0, virtual manufacturing approaches are increasingly used to optimize production and inspection processes. For desktop three-dimensional (3D) printers, calibration drift caused by mechanical wear can introduce positioning inaccuracies. Traditional calibration approaches rely on mechanical dial gauges and repeated manual measurements. This study presents a machine vision-based virtual encoder for XY-axis calibration. The system incorporates a custom chessboard marker, Edge Drawing Lines (EDLines) detection, sub-pixel corner refinement, and iterative Perspective-n-Point (PnP) pose estimation to obtain physical position measurements. These measurements support calibration of steps-per-millimeter settings, reversal-related lost motion, and cross-axis drift. Experimental comparisons with a dial gauge yielded mean absolute displacement discrepancies of 4.96 μm and 7.81 μm for the X and Y axes, respectively. After compensation, mean positioning errors decreased by 84.6% and 90.2%, reaching 7.1 μm and 5.4 μm, respectively. Elliptical trajectory evaluation showed a 54.24% reduction in total geometric path root-mean-square error (RMSE), from 100.9 μm to 46.17 μm. These results demonstrated improved linear-axis positioning and coordinated motion performance under the tested conditions.**

***Index Terms*—Virtual Encoder, Machine Vision, 3D Printer Calibration, Edge Drawing Lines, Chessboard Marker.**

I. Introduction

THE integration of physical and digital systems in modern manufacturing has supported virtual manufacturing approaches based on sensor data and computational models [1]. In material-extrusion 3D printing, dimensional and geometric accuracy are influenced by machine kinematics and material processing. Previous studies investigated kinematic error compensation in desktop 3D printers [2] and the effects of heat treatment on the dimensional accuracy of printed thermoplastic parts [3]. Kinematic sensitivity and uncertainty can affect positioning performance [4], while transmission backlash can introduce lost motion during direction reversal [5]. Identifying and compensating these deviations are therefore important for improving machine positioning performance.

Conventional displacement verification uses reference instruments such as mechanical dial gauges to compare measured motion with commanded displacement [6]. Manual instrument positioning and readout require repeated operator involvement, particularly when measurements are performed along multiple axes. Automated calibration approaches have been investigated using inductive sensing for multi-material fused filament fabrication printers [7] and machine vision for dual-nozzle spatial calibration [8]. The latter combines color segmentation, circle detection, and contour extraction to locate the nozzle positions. Previous work also investigated errors in circular trajectories generated by a 3D printer [9], providing context for coordinated-axis motion evaluation. These studies addressed nozzle calibration and trajectory error characterization. The present study combines image-based displacement measurement with parameter identification and motion compensation, and evaluates the resulting performance in both single-axis and coordinated motion.

Camera-based measurement enables non-contact displacement observation through image processing. Its performance can be evaluated against independent displacement sensors and is affected by environmental conditions such as illumination [10]. For checkerboard-based measurement, reliable sub-pixel corner localization is an important component of the vision measurement pipeline [11]. Machine calibration further requires the measured motion to be related to error modeling and compensation [12]. Accordingly, the proposed system is evaluated using reference displacement measurements and motion tests before and after compensation, in addition to reprojection error assessment.

This study proposes a machine vision-based virtual encoder for XY-axis calibration of a desktop 3D printer. The system combines a planar chessboard marker, EDLines-based feature extraction [13], gradient-based least-squares sub-pixel corner refinement, and iterative Perspective-n-Point (PnP) pose estimation [14]. The estimated marker positions are used to characterize positioning scale, reversal-related lost motion, and cross-axis drift. The resulting parameters support steps-per-millimeter adjustment and displacement-command compensation. The scope is restricted to XY motion; Z-axis calibration is not evaluated in this study.

The main contributions are the integration of vision-based position measurement with an axis calibration procedure, the identification and compensation of multiple motion-error components, and experimental evaluation against a dial gauge

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reference and along coordinated elliptical paths. Measurement discrepancies and uncertainty are considered alongside the changes in machine motion performance before and after calibration.

The remainder of this paper is organized as follows. Section II presents the preliminaries and calibration problem formulation. Section III describes the proposed machine vision-based virtual encoder and compensation scheme. Section IV presents the experimental setup, evaluation procedures, results, and discussion. Section V concludes the paper.

II. Problem Formulation and Preliminaries

This section introduces the mechanical motion errors and camera projection model used in XY-axis calibration. The calibration considers positioning scale, reversal-related lost motion, and the geometric relationship between the machine axes.

A. Positioning Scale and Steps-per-Millimeter

For a belt-driven axis, the nominal steps-per-millimeter setting depends on the motor step count, microstepping configuration, belt pitch, and pulley tooth count:

$$s_{\mathrm{nom}} = \frac{N_{\mathrm{step}}\mu}{p_b N_{\mathrm{tooth}}}, \tag{1}$$

where N_{step} is the number of full motor steps per revolution, μ is the microstepping factor, p_b is the belt pitch in millimeters, and N_{tooth} is the pulley tooth count. The product $p_b N_{\mathrm{tooth}}$ gives the nominal linear travel per pulley revolution.

A mismatch between the configured motion scale and the effective mechanical transmission produces a difference between the commanded displacement D_{cmd} and measured displacement D_{act}. Under a local linear approximation, the scale factor and associated displacement error are

$$\begin{aligned} \alpha &= \frac{D_{\mathrm{act}}}{D_{\mathrm{cmd}}}, \\ e_{\mathrm{scale}} &= D_{\mathrm{act}} - D_{\mathrm{cmd}} = (\alpha - 1) D_{\mathrm{cmd}}, \end{aligned} \tag{2}$$

where $D_{\mathrm{cmd}} \neq 0$. The dimensionless factor $\alpha = 1$ represents an ideal positioning scale. Measurements with a consistent approach direction are used to limit the influence of reversal-related lost motion during scale identification.

B. Backlash and Axis Non-Orthogonality

Backlash is reversal-related lost motion caused by clearance or slack in the transmission mechanism [5]. Following a direction reversal, part of the commanded actuator movement may be absorbed before the axis responds. Under a simplified lost-motion model,

$$D_{\mathrm{out}} = \max(0, D_{\mathrm{in}} - B_A), \tag{3}$$

where D_{in} and D_{out} are the commanded and resulting displacement magnitudes after reversal, respectively, and B_A is the lost-motion distance for axis $A \in \{X, Y\}$. This model assumes engagement in the previous movement direction and treats positioning scale separately.

Axis non-orthogonality describes a deviation from the nominal perpendicular relationship between the X and Y movement directions. For nonzero displacement vectors $\mathbf{v}_X$ and $\mathbf{v}_Y$ measured during separate positive-direction axis movements in a common frame, their internal angle is

$$\theta_{XY} = \cos^{-1}\left(\frac{\mathbf{v}_X^{\mathsf{T}}\mathbf{v}_Y}{\|\mathbf{v}_X\|_2\|\mathbf{v}_Y\|_2}\right). \tag{4}$$

The deviation from orthogonality is

$$\delta_{\mathrm{orth}} = \theta_{XY} - \frac{\pi}{2}, \tag{5}$$

where $\delta_{\mathrm{orth}} = 0$ indicates perpendicular axes.

A common rotation of the machine axes relative to the camera can introduce cross-axis components without changing the angle between the axes. Therefore, cross-axis drift measured in the camera frame must be distinguished from mechanical non-orthogonality.

C. Coordinate Frames and Camera Projection Model

The board frame defines the physical coordinates of the chessboard corners, the camera frame describes their positions relative to the camera, and the machine frame defines the nominal motion axes. Image coordinates (u, v) are expressed in pixels, increasing to the right and downward, respectively, from the upper-left image corner.

For a board point $\mathbf{P}_j = [X_j, Y_j, Z_j]^{\mathsf{T}}$, its coordinates in the camera frame are

$$\mathbf{P}_{c,ij} = R_i \mathbf{P}_j + t_i, \tag{6}$$

where R_i and t_i are the board rotation and translation for image i. For the planar chessboard, $Z_j = 0$. The translation has the same length unit as the board coordinates.

The ideal pinhole projection is

$$Z_{c,ij}\begin{bmatrix} u_{ij}^{(0)} \\ v_{ij}^{(0)} \\ 1 \end{bmatrix} = K(R_i \mathbf{P}_j + t_i), \tag{7}$$

with

$$K = \begin{bmatrix} f_u & 0 & c_u \\ 0 & f_v & c_v \\ 0 & 0 & 1 \end{bmatrix}.$$

Here, $Z_{c,ij}$ is the point depth in the camera frame, f_u and f_v are focal lengths in pixels, and (c_u, c_v) is the principal point. The superscript (0) denotes undistorted image coordinates, and zero pixel-axis skew is assumed.

Using normalized coordinates $x = X_c/Z_c$ and $y = Y_c/Z_c$, radial and tangential distortion are represented as

$$\begin{aligned} x_d &= x\left(1 + \sum_{\ell=1}^{L_r} k_\ell r^{2\ell}\right) + 2p_1 xy + p_2(r^2 + 2x^2), \\ y_d &= y\left(1 + \sum_{\ell=1}^{L_r} k_\ell r^{2\ell}\right) + p_1(r^2 + 2y^2) + 2p_2 xy, \end{aligned} \tag{8}$$

where $r^2 = x^2 + y^2$, k_ℓ are radial distortion coefficients, p_1 and p_2 are tangential distortion coefficients, and L_r is the number

of retained radial terms. The corresponding pixel coordinates are

$$u = f_u x_d + c_u, \qquad v = f_v y_d + c_v.$$

The complete projection is denoted by $\mathbf{c}_{ij} = \pi(\mathbf{P}_j; K, D, R_i, t_i)$, where D contains the distortion coefficients.

Pose estimation provides the board translation relative to the camera. Differences between translation estimates describe relative motion in the camera frame when the camera orientation remains fixed. Their correspondence to machine-axis displacement depends on the mounting configuration and coordinate alignment.

D. Calibration Problem Formulation

The calibration problem is to identify positioning scale, reversal-related lost motion, and cross-axis drift from measured XY movements. The camera intrinsic matrix K and distortion coefficients D are obtained through camera calibration and treated as fixed during machine parameter identification.

Initial, forward, and reversal position measurements are used to estimate the motion-error parameters for each axis. These estimates support adjustment of the steps-per-millimeter setting and compensation of displacement commands. The identification and compensation procedures are presented in Section III.

The objective is to reduce positioning and trajectory deviations relative to the baseline configuration. Measurement performance and compensation effectiveness are evaluated using a dial gauge reference and linear and coordinated XY motion tests, as described in Section IV.

III. Proposed Machine Vision-Based Virtual Encoder System

The proposed system uses chessboard images to estimate relative displacement and identify calibration parameters for the printer's X and Y axes. The measurement procedure comprises EDLines-based corner extraction, sub-pixel refinement, homography-based corner ordering, and iterative pose estimation using the camera model introduced in Section II. The ExtractPosition function in Algorithm 1 provides the position measurements, while the CalibrateAxis function in Algorithm 2 uses these measurements to estimate the parameters for positioning scale, reversal-related lost motion, and cross-axis compensation.

A. System Architecture and Marker Selection

The proposed system comprises a camera, a planar chessboard marker, a host computer for image processing, and the printer controller. The camera observes the marker during machine motion, and the host computer processes the captured images to estimate relative displacement. These measurements are subsequently used to identify calibration parameters and calculate motion compensation.

Following the marker-based measurement approach in [15], this study uses a chessboard instead of an ArUco marker. Chessboard intersections provide corner features suitable for sub-pixel localization using EDLines-based detection [13]. The selected board contains 8×8 squares and provides 7×7 internal corners with known physical coordinates for camera calibration and pose estimation.

ArUco markers provide encoded identities, while ChArUco boards combine marker identification with chessboard corners. The intended configuration uses a single board with the complete target grid visible, so encoded identification is not required. Three asymmetrically arranged circular anchors are included to support selection and ordering of the target corners.

The ordered corner coordinates are processed using iterative SolvePnP [14], based on the camera model introduced in Section II. The resulting position measurements are used by the parameter identification and compensation procedure described later in this section.

B. Sub-Pixel Feature Extraction Pipeline

The feature extraction pipeline operates on the raw grayscale image I_{gray} to preserve the local intensity gradients used for corner refinement. Contrast enhancement and smoothing are restricted to the anchor detection stage described in Section III-C.

EDLines detects line segments associated with the chessboard boundaries. Candidate intersections are screened using segment lengths, intersection proximity, and angular consistency to obtain initial corner estimates. Each estimate is then refined using a gradient-based least-squares method.

For an initial corner estimate $\mathbf{p}_0$, the refined position is obtained by minimizing the squared local gradient-orthogonality residuals:

$$\mathbf{p}^* = \arg\min_{\mathbf{p}\in\mathbb{R}^2} \sum_{\mathbf{q}_i \in N(\mathbf{p}_0)} \left[\mathbf{g}_i^{\mathsf{T}}(\mathbf{q}_i - \mathbf{p})\right]^2, \tag{9}$$

where $\mathbf{p}^*$ is the refined sub-pixel corner position, $\mathbf{p}_0 = [u_0, v_0]^{\mathsf{T}}$ is the initial estimate, and $N(\mathbf{p}_0)$ is its local pixel neighborhood. The vector $\mathbf{q}_i = [u_i, v_i]^{\mathsf{T}}$ denotes a neighboring pixel position, while $\mathbf{p} = [u, v]^{\mathsf{T}}$ is the candidate corner position. The image coordinates follow the convention introduced in Section II.

The local image intensity gradient at $\mathbf{q}_i$ is defined as

$$\mathbf{g}_i = \nabla I_{\text{gray}}(\mathbf{q}_i) = \left.\begin{bmatrix} \dfrac{\partial I_{\text{gray}}}{\partial u} \\ \dfrac{\partial I_{\text{gray}}}{\partial v} \end{bmatrix}\right|_{(u,v)=(u_i,v_i)},$$

where ∇ denotes the spatial gradient operator. The components of $\mathbf{g}_i$ describe local intensity changes along the horizontal and vertical directions, respectively.

The residual $\mathbf{g}_i^{\mathsf{T}}(\mathbf{q}_i - \mathbf{p})$ expresses the local orthogonality condition between the image gradient and the displacement from the candidate corner to a neighboring pixel. For a fixed neighborhood, setting the derivative of (9) with respect to $\mathbf{p}$ to zero gives

$$\mathbf{p}^* = \left(\sum_{\mathbf{q}_i \in N(\mathbf{p}_0)} \mathbf{g}_i \mathbf{g}_i^{\mathsf{T}}\right)^{-1} \left(\sum_{\mathbf{q}_i \in N(\mathbf{p}_0)} \mathbf{g}_i \mathbf{g}_i^{\mathsf{T}} \mathbf{q}_i\right). \tag{10}$$

This solution requires the 2×2 matrix inside the inverse to be nonsingular. The refined corner coordinates are subsequently used for grid ordering and pose estimation.

C. Homography Alignment and Corner Sorting

The extracted sub-pixel corners are arranged into a consistent 7×7 grid with the aid of the three circular anchors on the marker. Image enhancement is restricted to anchor detection, while the original corner coordinates obtained from the raw grayscale image are retained.

Contrast-limited adaptive histogram equalization (CLAHE) improves local contrast around the anchor patterns, while Gaussian filtering reduces noise in the anchor detection stage. The 5×5 Gaussian kernel is obtained by sampling and normalizing

$$G_2(x, y) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{x^2 + y^2}{2\sigma^2}\right), \tag{11}$$

where x and y are local pixel offsets from the kernel center, and σ is the Gaussian standard deviation in pixels. These offsets are distinct from the normalized camera coordinates defined in Section II.

Following anchor detection, candidate corners are filtered to select the target grid. A homography H is estimated using the four outermost grid corners and their corresponding ideal grid coordinates. The mapping from the ideal grid to the image is expressed as

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = H \begin{bmatrix} a \\ b \\ 1 \end{bmatrix}, \tag{12}$$

where $a, b \in \{0, \ldots, 6\}$ denote the column and row indices of an ideal grid point, (u, v) is its projected image location, and λ is a homogeneous scale factor. The 3×3 matrix H is defined up to scale.

The projected grid locations are used to associate detected corners with their grid indices and arrange them in row-major order. This procedure establishes consistent point correspondences without replacing the original sub-pixel corner coordinates, which are subsequently used for camera calibration and pose estimation.

D. Virtual Encoder Position Extraction

The ordered sub-pixel corners and their corresponding physical board coordinates are supplied to an iterative SolvePnP solver [14]. Using the projection model introduced in Section II, the solver estimates the board rotation $\hat{R}_i$ and translation $\hat{t}_i$ relative to the camera.

The estimated pose is evaluated using the reprojection root-mean-square error:

$$\epsilon_i = \sqrt{\frac{1}{m} \sum_{j=1}^{m} \|\mathbf{c}_{ij} - \hat{\mathbf{c}}_{ij}\|_2^2}, \tag{13}$$

where $m = 49$ is the number of corners, $\mathbf{c}_{ij}$ is the observed position of corner j in image i, and

$$\hat{\mathbf{c}}_{ij} = \pi(\mathbf{P}_j; K, D, \hat{R}_i, \hat{t}_i)$$

Algorithm 1 EXTRACTPOSITION Function

Require: Image I_i, calibrated parameters K, D, board points $\{\mathbf{P}_j\}_{j=1}^{49}$, threshold ϵ_{th}
Ensure: Position $\mathbf{p}_i$ and error ϵ_i, or (Null, ∞)
1: **function** EXTRACTPOSITION($I_i, K, D, \{\mathbf{P}_j\}, \epsilon_{\text{th}}$)
2: **if** I_i is invalid or empty **then**
3: **return** Null, ∞
4: **end if**
5: Convert I_i to grayscale image I_{gray}.
6: Extract and refine corners C_{sub} using (9) and (10).
7: Order C_{sub} into a 7×7 grid to obtain C_{sorted}, as described in Section III-C.
8: **if** C_{sorted} is invalid or does not contain 49 ordered corners **then**
9: **return** Null, ∞
10: **end if**
11: Estimate $(\hat{R}_i, \hat{t}_i)$ using iterative SolvePnP with C_{sorted}, $\{\mathbf{P}_j\}$, K, and D.
12: **if** pose estimation fails **then**
13: **return** Null, ∞
14: **end if**
15: Reproject the board points to obtain $\hat{\mathbf{c}}_{ij}$ for $j = 1, \ldots, 49$.
16: Calculate ϵ_i using (13).
17: **if** $\epsilon_i > \epsilon_{\text{th}}$ **then**
18: **return** Null, ∞
19: **end if**
20: Extract $\mathbf{p}_i$ using (14).
21: **return** $\mathbf{p}_i, \epsilon_i$
22: **end function**

is its reprojected image position. The error ϵ_i and acceptance threshold ϵ_{th} are expressed in pixels.

For an accepted pose, the position readout is extracted from the translation vector:

$$\mathbf{p}_i = \begin{bmatrix} \hat{T}_{x,i} \\ \hat{T}_{y,i} \end{bmatrix}, \qquad \hat{t}_i = \begin{bmatrix} \hat{T}_{x,i} \\ \hat{T}_{y,i} \\ \hat{T}_{z,i} \end{bmatrix}. \tag{14}$$

The components $\hat{T}_{x,i}$, $\hat{T}_{y,i}$, and $\hat{T}_{z,i}$ describe translation along the camera coordinate axes. The position has the same length unit as the physical board coordinates. Relative displacement is subsequently calculated as $\Delta\mathbf{p}_i = \mathbf{p}_i - \mathbf{p}_{\text{ref}}$, where $\mathbf{p}_{\text{ref}}$ is obtained from a reference image.

Algorithm 1 describes the EXTRACTPOSITION function. Line 1 defines its inputs. Lines 2–4 reject invalid images. Lines 5–7 convert the image to grayscale, refine the corners using (9) and (10), and establish their grid order using Section III-C. Lines 8–10 reject invalid corner sets. Lines 11–14 estimate the pose and handle estimation failure. Lines 15–19 calculate the reprojection error using (13) and reject estimates above the threshold. Lines 20–22 extract the position using (14), return the result, and complete the function.

E. Kinematic Parameter Identification and Compensation

For each active axis $A \in \{X, Y\}$, initial, forward, and reversal images are processed using EXTRACTPOSITION in Algorithm 1. The resulting position components are interpreted along the active and passive machine directions after establishing their correspondence with the camera coordinate frame.

For measurement cycle i, the displacement and error metrics are calculated as

$$\begin{aligned} d_i &= p_{\mathrm{fwd,act},i} - p_{\mathrm{init,act},i}, \\ c_i &= p_{\mathrm{fwd,pas},i} - p_{\mathrm{init,pas},i}, \\ B_i &= \left| p_{\mathrm{rev,act},i} - p_{\mathrm{init,act},i} \right|, \\ \eta_i &= \frac{c_i}{d_i}. \end{aligned} \tag{15}$$

Here, d_i and c_i are the active-axis and passive-axis displacements, respectively. The quantity B_i estimates reversal-related lost motion under the prescribed movement sequence, and η_i describes cross-axis drift in the selected measurement frame. Only valid position triplets satisfying $|d_i| > d_{\min}$ are retained, where $d_{\min} > 0$ prevents division by a near-zero displacement.

The means of the n_v accepted measurements are

$$\begin{bmatrix} \bar{d} \\ \bar{B} \\ \bar{\eta} \end{bmatrix} = \frac{1}{n_v} \sum_{i=1}^{n_v} \begin{bmatrix} d_i \\ B_i \\ \eta_i \end{bmatrix}. \tag{16}$$

The calibration parameters are obtained as

$$\begin{aligned} s_{\mathrm{new}} &= s_{\mathrm{curr}} \frac{D_{\mathrm{cmd}}}{\bar{d}}, \\ B_{\mathrm{comp}} &= \bar{B}, \\ \eta_{\mathrm{comp}} &= \bar{\eta}. \end{aligned} \tag{17}$$

The current and updated steps-per-millimeter settings are denoted by s_{curr} and s_{new}, respectively. The commanded displacement D_{cmd} and measured displacement $\bar{d}$ follow the same sign convention. The resulting parameter set is $\Theta_A = (s_{\mathrm{new}}, B_{\mathrm{comp}}, \eta_{\mathrm{comp}})$.

Algorithm 2 describes the CALIBRATEAXIS function. Line 1 defines the function, and line 2 initializes the accepted measurement set. Lines 3–4 process each image triplet using EXTRACTPOSITION. Lines 5–7 calculate and retain the metrics in (15) when the positions are valid and the active-axis displacement is sufficient. Lines 8–10 stop collection at $N_{\max}$ accepted measurements, and line 11 completes the loop. Lines 12–14 reject insufficient data or a near-zero mean displacement. Lines 15–16 calculate the means and calibration parameters using (16) and (17). Lines 17–18 return the parameters and complete the function.

The camera parameters, board coordinates, and thresholds listed under **Require** are supplied as fixed settings to the function. The quantities $N_{\min}$ and $N_{\max}$ specify the minimum required and maximum collected numbers of valid measurements, respectively.

The updated steps-per-millimeter setting is applied using the axis-specific `M92` command. For a single-axis commanded displacement $\Delta\mathbf{q} = [\Delta x, \Delta y]^{\mathsf{T}}$, cross-axis compensation is expressed as

$$\Delta\mathbf{q}_{\mathrm{skew}} = \begin{cases} \begin{bmatrix} \Delta x \\ \Delta y - \eta_{\mathrm{comp}} \Delta x \end{bmatrix}, & A = X, \\ \begin{bmatrix} \Delta x - \eta_{\mathrm{comp}} \Delta y \\ \Delta y \end{bmatrix}, & A = Y. \end{cases} \tag{18}$$

The reversal-dependent correction is

$$\Delta\mathbf{q}_{\mathrm{comp}} = \Delta\mathbf{q}_{\mathrm{skew}} + b B_{\mathrm{comp}} \operatorname{sgn}(\Delta q_A) \mathbf{e}_A, \tag{19}$$

Algorithm 2 CALIBRATEAXIS Function

Require: Image triplets $\mathcal{S}$ for axis A, camera parameters K, D, board points $\{\mathbf{P}_j\}$, D_{cmd}, s_{curr}, ϵ_{th}, $d_{\min} > 0$, $1 \le N_{\min} \le N_{\max}$
Ensure: Calibration parameters Θ_A, or Null

1: **function** CALIBRATEAXIS($\mathcal{S}, A, D_{\mathrm{cmd}}, s_{\mathrm{curr}}$)
2: Initialize an empty measurement set $\mathcal{M}$.
3: **for** each image triplet in $\mathcal{S}$ **do**
4: Obtain the three positions using EXTRACTPOSITION with $K, D, \{\mathbf{P}_j\}, \epsilon_{\mathrm{th}}$.
5: **if** all positions are valid and the active-axis displacement satisfies $|d_i| > d_{\min}$ **then**
6: Calculate and store (d_i, B_i, η_i) in $\mathcal{M}$ using (15).
7: **end if**
8: **if** $|\mathcal{M}| \ge N_{\max}$ **then**
9: **break**
10: **end if**
11: **end for**
12: **if** $|\mathcal{M}| < N_{\min}$ or $|\operatorname{Mean}_{\mathcal{M}}(d_i)| \le d_{\min}$ **then**
13: **return** Null
14: **end if**
15: Calculate $\bar{d}, \bar{B}, \bar{\eta}$ using (16).
16: Calculate Θ_A using (17).
17: **return** Θ_A
18: **end function**

where Δq_A is the commanded active-axis displacement, $\mathbf{e}_X = [1, 0]^{\mathsf{T}}$, and $\mathbf{e}_Y = [0, 1]^{\mathsf{T}}$. The flag b equals one when the active axis reverses direction and zero otherwise. When $b = 0$, only the cross-axis correction is applied.

IV. EXPERIMENTAL RESULTS AND DISCUSSION

This section presents the experimental setup, measurement procedures, and evaluation results of the proposed virtual encoder. Camera calibration and reference measurements are first described to characterize measurement performance and uncertainty. Linear-axis and elliptical trajectory results are then compared before and after machine calibration to assess compensation effectiveness.

A. Experimental Setup and Camera Calibration

The proposed system was evaluated using a Creality CR-10 V3 printer with an XZ-head/Y-bed configuration. An OV9821 camera module was configured to capture images at 1280×720 pixels and 120 frames per second. The camera was attached to the X-carriage using a custom PLA mount, with a working distance of approximately 0.60 m between the lens and the marker during acquisition. A planar chessboard marker was affixed to the Y-axis print bed so that the camera observed relative motion between the carriage and the bed. The experimental configuration is shown in Fig. 1.

The marker contained 8×8 squares with a nominal side length of 20 mm, providing 49 internal corners. Three circular anchors with a radius of 5 mm were included to support corner selection and ordering.

Before displacement measurement, the camera was calibrated using a dataset of ten board images captured at different poses. Corners were extracted and refined using (9) and (10), then ordered using the procedure in Section III-C. Images with a complete and successfully ordered 7×7 grid were retained for calibration. A representative result of feature extraction and corner ordering is shown in Fig. 2.

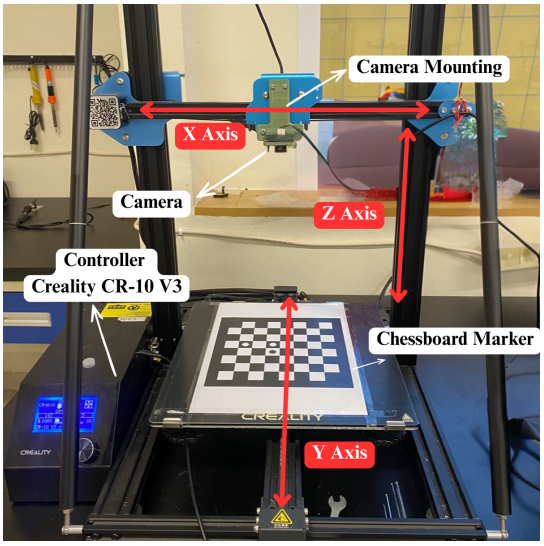


Fig. 1: Experimental configuration of the camera-based virtual encoder.

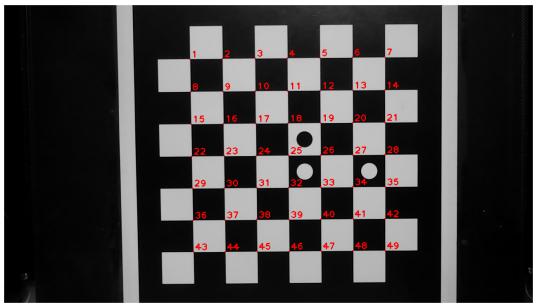


Fig. 2: Detected and ordered chessboard corners in a representative calibration image.

The physical board coordinates were paired with the detected image coordinates to estimate the intrinsic matrix K, distortion coefficients D, and board pose in each calibration image. The parameters were obtained by minimizing the sum of squared reprojection errors:

$$\underset{K,D,\{R_i,t_i\}_{i=1}^{n}}{\text{minimize}} \quad \sum_{i=1}^{n}\sum_{j=1}^{m} \|\mathbf{c}_{ij} - \pi(\mathbf{P}_j; K, D, R_i, t_i)\|_2^2 , \quad (20)$$

where n is the number of retained calibration images and $m = 49$ is the number of corners per image. The vector $\mathbf{c}_{ij}$ denotes the detected sub-pixel position of corner j in image i. The corresponding board coordinates $\mathbf{P}_j$ were expressed in millimeters. The projection function $\pi(\cdot)$ follows the camera model described in Section II.

The estimated intrinsic matrix and distortion coefficient vector were

$$K = \begin{bmatrix} 2314.64 & 0 & 637.08 \\ 0 & 2328.29 & 222.87 \\ 0 & 0 & 1 \end{bmatrix} \quad (21)$$

and

$$D = \begin{bmatrix} -0.0561 & 0.9595 & 0.0001 & 0.0004 & -5.0330 \end{bmatrix}, \quad (22)$$

respectively. The entries of D correspond to $[k_1, k_2, p_1, p_2, k_3]$, comprising three radial and two tangential distortion coefficients.

Calibration quality was evaluated using the overall reprojection RMSE:

$$e_{\text{cal}} = \sqrt{\frac{1}{nm}\sum_{i=1}^{n}\sum_{j=1}^{m} \left\| \mathbf{c}_{ij} - \pi(\mathbf{P}_j; K, D, \hat{R}_i, \hat{t}_i) \right\|_2^2}, \quad (23)$$

where the projection was evaluated using the estimated camera parameters and board poses. The resulting RMSE was expressed in pixels.

The previous calibration results reported a mean reprojection error of 0.1152 pixels, with a standard deviation of 0.0076 pixels. The calibrated K and D were subsequently used by ExtractPosition in Algorithm 1.

B. Measurement Procedure and Evaluation Metrics

Measurement performance was evaluated through reference displacement comparisons, single-axis movement tests, and elliptical trajectory tests. The motion tests were performed under baseline and post-calibration conditions using the same evaluation metrics.

For reference measurements, each axis was tested separately against a mechanical dial gauge. A commanded displacement of $D_{\text{cmd}} = 0.5$ mm was selected within the stated single-revolution range of 0.8 mm. Each measurement cycle comprised initial, forward, and reversal frames, as illustrated in Fig. 3.

The initial frame was captured at the starting reference position. The forward frame was captured after movement by $+D_{\text{cmd}}$, and the reversal frame was captured after movement by $-D_{\text{cmd}}$ back toward the nominal starting position. Before the next cycle, an additional movement of $-D_{\text{cmd}}$ followed by $+D_{\text{cmd}}$ established a consistent approach direction. A new initial frame was then captured.

Positions were extracted using ExtractPosition in Algorithm 1. Only image triplets with valid position estimates and reprojection errors satisfying $\epsilon_i \leq \epsilon_{\text{th}}$ for all three frames were retained. Encoder displacements relative to the initial position were compared with the corresponding dial gauge readings.

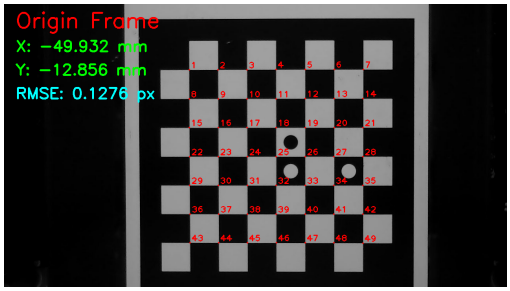
For each movement condition, the mean absolute measurement discrepancy was calculated as

$$e_{\text{MAE}} = \frac{1}{N_{\text{valid}}}\sum_{i=1}^{N_{\text{valid}}} |d_{\text{enc},i} - d_{\text{dial},i}| , \quad (24)$$

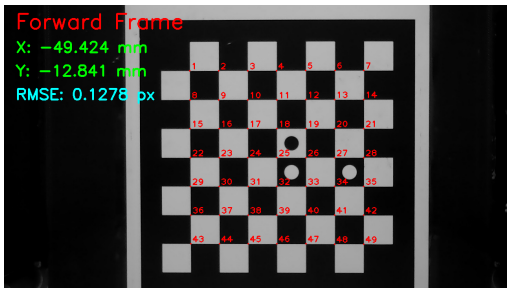
where N_{valid} is the number of accepted measurement cycles. The quantities $d_{\text{enc},i}$ and $d_{\text{dial},i}$ denote the encoder and reference displacement readouts, respectively. Both readouts were expressed in the same length unit and used the same direction convention.

Single-axis motion was evaluated using the same initial, forward, and reversal sequence. The signed positioning deviation was calculated as

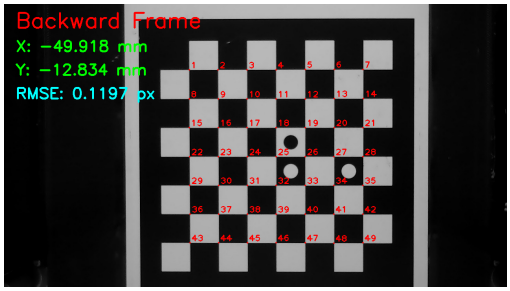
$$e_{\text{pos},i} = d_i - D_{\text{cmd}},$$



(a)



(b)



(c)

Fig. 3: Representative position extraction results for one active axis: (a) initial frame, (b) forward frame, and (c) reversal frame.

where d_i is the measured active-axis displacement. Reversal-related lost motion and cross-axis drift were evaluated using (15). The calibration parameters were obtained using Algorithm 2. The same movement sequence and error metrics were used before and after compensation.

Coordinated X–Y motion was evaluated along the commanded elliptical path shown in Fig. 4. The printer executed the path through G-code, and video frames were processed using ExtractPosition. Invalid position estimates and frames with reprojection errors exceeding ϵ_{th} were excluded.

For each trial, the measured trajectory was translated to align its first accepted position with the reference starting point:

$$\tilde{\mathbf{p}}_k = \mathbf{p}_k - \mathbf{p}_0 + \mathbf{r}_0, \tag{25}$$

where $\mathbf{p}_k$ is the extracted position, $\mathbf{p}_0$ is the first accepted position, and $\mathbf{r}_0$ is the reference starting point. Measured and reference positions were expressed in a common coordinate frame and in millimeters.

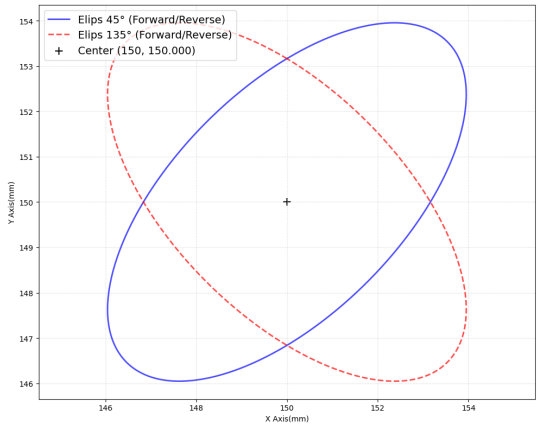


Fig. 4: Commanded elliptical trajectory for coordinated X–Y motion evaluation.

Each aligned position was associated with the nearest sampled reference point:

$$\begin{aligned} j_k^* &= \arg\min_j \|\tilde{\mathbf{p}}_k - \mathbf{r}_j\|_2, \\ \mathbf{e}_k &= \tilde{\mathbf{p}}_k - \mathbf{r}_{j_k^*}, \end{aligned} \tag{26}$$

where $\mathbf{r}_j$ is a sampled reference point and $\mathbf{e}_k = [e_{x,k}, e_{y,k}]^{\mathsf{T}}$ is the geometric deviation. A KD-tree was used for nearest-neighbor search.

For L accepted positions, the trajectory deviation metrics were calculated as

$$\begin{aligned} E_x &= 10^3 \sqrt{\frac{1}{L}\sum_{k=1}^{L} e_{x,k}^2}, \\ E_y &= 10^3 \sqrt{\frac{1}{L}\sum_{k=1}^{L} e_{y,k}^2}, \\ E_{\text{tot}} &= 10^3 \sqrt{\frac{1}{L}\sum_{k=1}^{L} \|\mathbf{e}_k\|_2^2}. \end{aligned} \tag{27}$$

The factor 10^3 converts millimeters to micrometers. These metrics quantify geometric deviation from the sampled reference path. They do not quantify motion delay or time-synchronized tracking error. Trial-wise RMSE values were summarized across repeated tests and compared between baseline and post-calibration conditions.

C. Measurement Discrepancy and Uncertainty

The virtual encoder displacement readouts were compared with the dial gauge measurements using (24). The measurement comparison for the forward and reversal conditions is shown in Fig. 5. Evaluated with (24), the mean absolute discrepancies were 4.96 μm for the X-axis and 7.81 μm for the Y-axis. The corresponding direction-dependent mean offsets were 3.8 μm forward and 3.3 μm backward on the X-axis, and 8.5 μm forward and 4.0 μm backward on the Y-axis. Because these offsets reverse sign with the travel direction, they partially cancel when averaged over both directions,

giving net offsets of -0.2 μm and $+2.3$ μm, respectively. These values described agreement with the reference readings over the tested 0.5 mm displacement range.

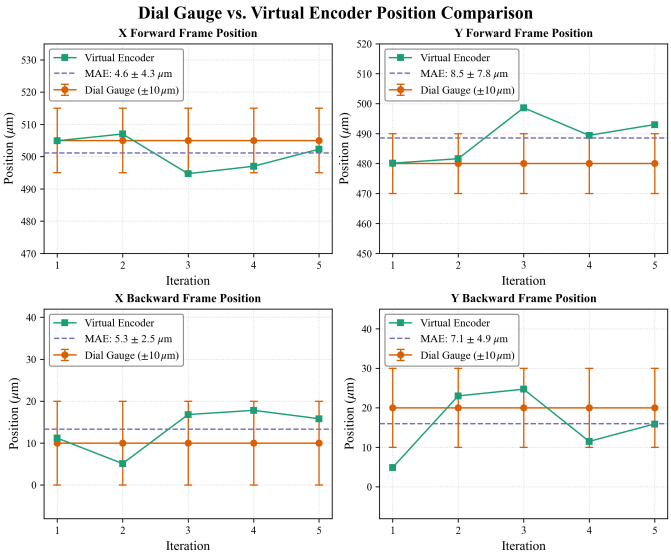


Fig. 5: Comparison of dial gauge and virtual encoder position readouts for the X and Y axes under forward and reversal conditions.

Stationary measurements were used to characterize fluctuations in the position readout, as shown in Fig. 6. The mean absolute readout deviations were 2.58 μm and 4.97 μm for the X and Y axes, and the standard deviations of the repeated stationary readout were 2.63 μm and 3.44 μm, respectively. These results characterized measurement stability under stationary conditions.

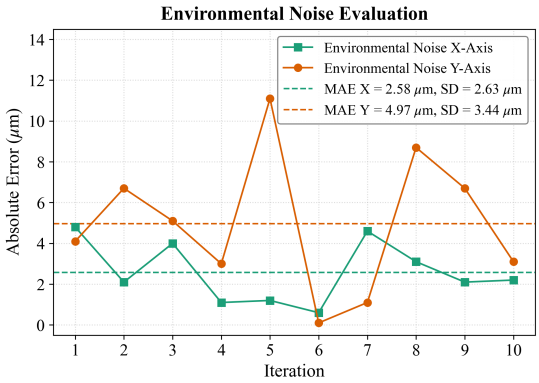


Fig. 6: Stationary position readout fluctuations for the X and Y axes.

An uncertainty budget was formulated using the ISO/IEC Guide 98-3 (GUM) framework. Assuming uncorrelated contributions expressed in displacement units, the combined standard uncertainty was calculated as

$$u_c = \sqrt{u_A^2 + u_{\text{dial}}^2}. \tag{28}$$

The Type A repeatability contribution was evaluated from $N = 5$ repeated measurements per travel direction as

$$u_A = \frac{\sigma}{\sqrt{N}},$$

where σ is the sample standard deviation of the repeated displacement measurements within a travel direction, pooled over the forward and reversal conditions. The stationary readout dispersion reported above is of the same order as this term and originates from the same readout noise; it was therefore characterized separately instead of being added as an independent contribution, so that the same dispersion is not counted twice. The sub-pixel reprojection residual was likewise retained as an indicator of model-fit quality rather than as an uncertainty contribution, because it is already reflected in the dispersion of the measured displacement.

For a reference instrument error interval $[-a, a]$ with a rectangular distribution, the standard uncertainty was calculated as

$$u_{\text{dial}} = \frac{a}{\sqrt{3}}.$$

The quantity a denotes the half-width of the assumed reference error interval and is distinct from the dial gauge resolution. Using $a = 10$ μm gave $u_{\text{dial}} = 5.77$ μm.

The expanded uncertainty was calculated using a coverage factor $k = 2$:

$$U = k u_c = 2u_c. \tag{29}$$

This coverage factor corresponds to approximately 95% coverage when the distributional and effective degrees-of-freedom assumptions are satisfied. The individual uncertainty contributions and the reported combined and expanded uncertainties are summarized in Table I.

TABLE I: Measurement Uncertainty Budget for the Virtual Encoder System

Source of Uncertainty	**Symbol**	**Distribution**	**X (μm)**	**Y (μm)**
Measurement Repeatability (Type A)	u_A	Normal	2.34	3.57
Reference Dial Gauge Tolerance	u_{dial}	Rectangular	5.77	5.77
Combined Standard Uncertainty	u_c	—	**6.23**	**6.79**
Expanded Uncertainty	U	—	**12.46**	**13.57**

Repeatability was evaluated using $N = 5$ repeated measurements per travel direction. The stationary readout dispersion is reported separately and is not added to the budget, so that the same readout noise is not counted twice. The sub-pixel reprojection residual is reported as a model-fit indicator, not as an uncertainty contribution. Expanded uncertainty used a coverage factor $k = 2$.

The dial gauge contribution was the largest individual component in the reported budget. The mean absolute encoder–reference discrepancies were smaller than the expanded uncertainties. However, this comparison did not establish coverage for every individual measurement or independently demonstrate absolute accuracy below 7 μm.

D. Linear-Axis Calibration Results

Baseline measurements showed mean positioning errors of 46.3 μm and 55.0 μm for the X and Y axes, respectively. The corresponding reversal-related errors were 12.4 μm and 24.4 μm. Single-axis movements also exhibited cross-axis

drift, with reported drift angles of -0.5344° for X displacement during Y motion and 0.5171° for Y displacement during X motion. These angles described motion relative to the measurement coordinate frame and did not independently establish mechanical axis non-orthogonality.

The calibration parameters were refined using Algorithm 2. Figures 7 and 8 compare positioning, cross-axis drift, and reversal-related errors across ten evaluation iterations for the X and Y axes, respectively, under baseline and calibrated conditions.

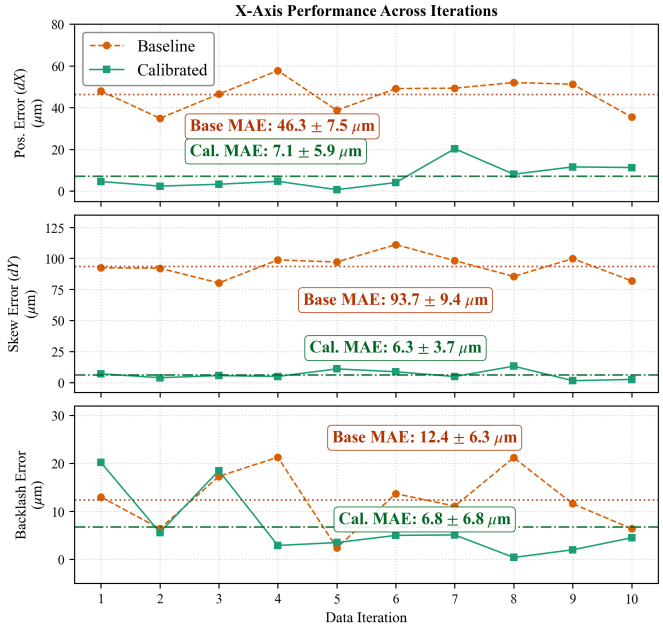


Fig. 7: X-axis positioning, cross-axis drift, and reversal-related errors across ten evaluation iterations under baseline and calibrated conditions.

Table II lists the final steps-per-millimeter settings, lost-motion compensation values, and cross-axis correction slopes. These values represented the applied compensation parameters rather than the residual errors after calibration.

TABLE II: Final Linear-Axis Calibration Parameters

Parameter	Description	Final Value
STEPX	X-axis steps-per-millimeter setting	79.72 steps/mm
STEPY	Y-axis steps-per-millimeter setting	80.40 steps/mm
BACKLASH_X	X-axis lost-motion compensation	13.3 μm
BACKLASH_Y	Y-axis lost-motion compensation	31.4 μm
SKEW_X_ON_Y	Parasitic X-drift during Y motion	-9.2 μm/mm
SKEW_Y_ON_X	Parasitic Y-drift during X motion	9.213 μm/mm

Cross-axis values denote the identified drift slopes used in (18). The listed parameters are compensation settings, not residual errors after calibration.

The mean absolute errors and percentage reductions are summarized in Fig. 9. After compensation, the X-axis positioning error decreased by 84.6% to 7.1 ± 5.9 μm. Its cross-axis drift and reversal-related errors decreased by 93.3% and 45.6%, reaching 6.3 ± 3.7 μm and 6.8 ± 6.8 μm, respectively. For the Y-axis, the positioning error decreased by 90.2% to 5.4 ± 5.4 μm. Cross-axis drift decreased by 94.5% to 5.0 ± 4.3 μm, while the reversal-related error decreased by 72.0% to 6.8 ± 5.2 μm.

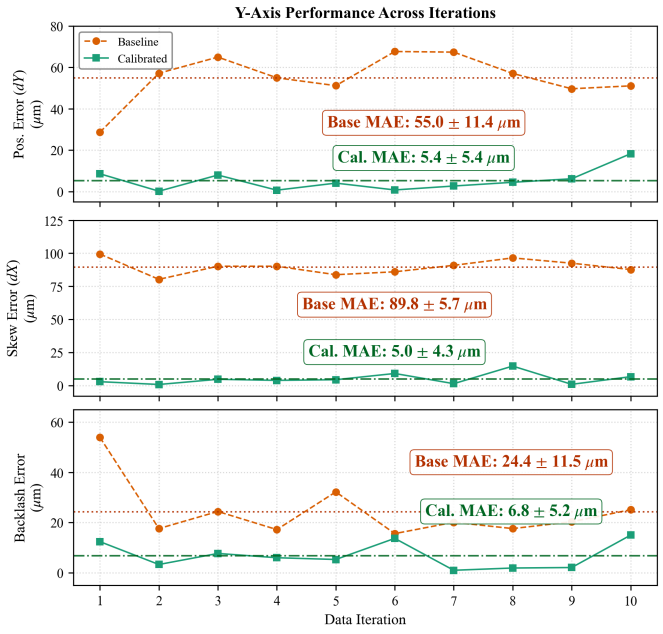


Fig. 8: Y-axis positioning, cross-axis drift, and reversal-related errors across ten evaluation iterations under baseline and calibrated conditions.

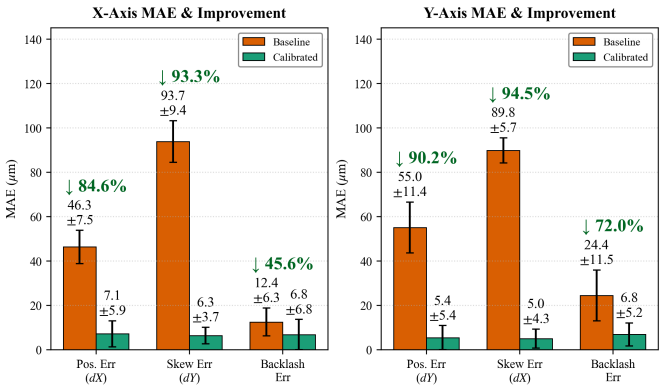


Fig. 9: Mean absolute errors and percentage reductions for positioning, cross-axis drift, and reversal-related error on the X and Y axes.

The results showed reductions in all evaluated linear-motion error components. Positioning and cross-axis drift exhibited larger percentage reductions than reversal-related error. The remaining deviations were interpreted alongside the measurement uncertainty reported in the preceding subsection. These results supported improved motion performance under the tested conditions.

E. Elliptical Trajectory Results

The elliptical trajectory tests evaluated coordinated X–Y motion under baseline and post-calibration conditions. Geometric deviations from the commanded reference path were

calculated using (27). The baseline total trajectory RMSE was 100.9 ± 7.4 μm.

After calibration, the X-axis trajectory RMSE decreased by 55.25% to 34.42 ± 1.70 μm, while the Y-axis RMSE decreased by 52.91% to 30.75 ± 1.37 μm. The total trajectory RMSE decreased by 54.24% to 46.17 ± 1.77 μm. These results showed that the calibration parameters obtained from single-axis measurements also reduced geometric deviations during coordinated motion.

The remaining trajectory deviations were greater than the residual errors observed in the linear-axis tests. This difference indicated that improved single-axis positioning did not eliminate deviations during continuous multi-axis movement. The trajectory results therefore complemented the linear-axis evaluation by assessing performance under coordinated motion.

Because the evaluation associated each measured position with the nearest sampled reference point, the reported RMSE values characterized geometric path deviation. They did not quantify motion delay or time-synchronized tracking error.

V. Conclusion

This study developed a low-cost, machine vision-based virtual encoder for XY-axis calibration of a 3D printer using a planar chessboard marker, EDLines-based corner extraction, and iterative PnP pose estimation. Measurement discrepancies relative to the dial gauge were 4.96 μm and 7.81 μm for the X and Y axes, respectively. After calibration, positioning errors decreased by 84.6% on the X-axis and 90.2% on the Y-axis, reaching 7.1 ± 5.9 μm and 5.4 ± 5.4 μm, respectively. Cross-axis drift and reversal-related errors also decreased. The total elliptical trajectory deviation decreased by 54.24%, from 100.9 μm to 46.17 ± 1.77 μm RMSE. These results demonstrated that vision-based measurement and software compensation improved the printer's motion performance under the tested conditions with a measurement hardware cost below USD 30.

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